

Activity Simulation from Signals

Tsuyoshi Okita
tsuyoshi.okita@gmail.com
Kyushu Institute of Technology
Kitakyushu, JAPAN

ABSTRACT

Sensor-based human activity recognition technology has been used for estimating human action based on the sensor data. In this paper, we propose a new paradigm to render the human activity on a screen instead of classifying the activity among the activity labels. We could built this mockup of a simulator, combining our previous translation tool between signals [2] with the motion rendering systems [3]. We faced two problems which decrease the simulation ability a lot. We proposed two algorithms to increase the performance of this simulator in this preliminary work.

KEYWORDS

activity recognition, simulation, neural rendering

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1 ACTIVITY SIMULATION

In traditional activity recognition systems, the targeted activity labels were often limited less than 10 or 20. However, recently this number of activity labels has been increased. Multiple sensors of smartphone sensors attached with human are often used for activity recognition systems in various situations. Even the motion capture systems of around 70 markers / sensors in the case of Helen Heyes marker set are used for this purpose. A system involving more than 50 types of complex activity / behavior recognition labels, this situation is quite different from the simple activity recognition sinarios in the past. There would be various approaches for this. One way is to make the traditional simple labels complex or hierarchical. The other way to handle this complex activity / behavior recognition case is to decompose the target, detecting interactions between multiple peoples and/or interactions between people and objects. There would be various strategies which can be taken along this line.

In order to see this situation from another angle to avoid this limitation, we propose a new realistic and affordable strategy to tackle this problem. Based on the signals, we converted back in

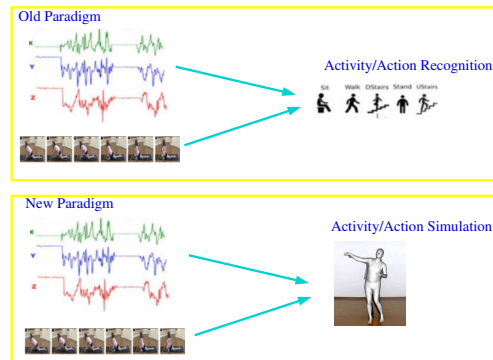


Figure 1: Activity Recognition and Activity Simulation

the form of 3D humans. That is, we can see the 3D motion of humanbeings directly on a simulation systems in which we are supposed to have collected sensor data. Therefore, we call this activity simulator. We have advantage in this simulator for complex activities. The simple activities such as walk or jump can be easily discriminated by the traditional activity recognition systems as well as this simulator. However, the complex activities, such as the lifting of the patient by the nurse or the injection activity by the nurse to the patient, are quite difficult to discriminate. This is since they looks alike in the signal level but the varieties of these activities in each category increases a lot. In this reason, the recognition rate by the traditional activity recognition systems is decreased. Even in such a case, the 3D simulation can be effective for discriminating these since the judgement is done by humans.

In our previous works, we develop the conversion tool between multimodal signals (e.g. the smartphone signals into the motion capture signals, or the video signals into the smartphone signals) [2]. After converted using this converter / translation of signals, we applied them to render the humanbeings. It is noted that we develop this simulator in a batch mode rather than online mode (real time). In other words, we develop this simulator that can see the data in the small future as well as the small past.

2 PROBLEMS

We could build the system quite smoothly using FrankMocap [3] combined with multimodal signal translation [2]. It can render the 3D human quite effectively based on the input signals. In the following algorithms, we first translate the signals into images or skeleton, then apply to FrankMocap. Therefore, the algorithm relies on the object detection method operating on image.

However, we faced with two small problems if used it as a simulator.

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- The number of 3D human drawn suddenly generates only a few frames and suddenly disappears, and
- the height of the frame is unstable which causes the sudden change of the height of the rendered human.

These two problems yields quite unnatural simulator if we look at them in our commonsense. These are clearly the problem of rendering algorithm by the FrankMocap. Therefore, in order to overcome these problems, we modified the algorithm using the following algorithms.

Although humans do not move, it is often the detected bounding box tends to move slightly if frames progress. Let us say a human raises his hand, the man does not move but only his hand moves. However, since the bounding boxes in the frame which captures a human is not fixed but slightly moves around if the frames go forward. Similarly, if the length of the skeleton detection chain changes, the size of detected bounding-box moves around if frames progress. This makes the length of the chain moves around. Therefore, the height of the simulated person changes suddenly. Therefore, we call these the first problem and makes the change of the size of bounding-box stable. We deploy the algorithm in the following.

- (1) We detect a situation where a human moves suddenly and significantly on the screen in continuous frames.
- (2) Unless there is a situation in which a human suddenly moves significantly on the screen, we interpolate the previous and the next frames as the current frame. We generate a 3D human based on this bounding box.
- (3) If there is a situation where a human moves suddenly and significantly on the screen, we do not use the previous frame and extrapolate the next frame. We generate a 3D human based on this bounding box.
- (4) Return to (1).

Suppose that the human is detected in some frame and suddenly disappeared in the next frame. This phenomenon occurs quite frequently since the detection of bounding box immediately generates the skelton, which renders the human. We consider this as the detection noise of human. Therefore we let them not draw since they are ghost.

- (1) If we detect the sudden generation of the humans based on the sudden generation of the bounding box, we do not render 3D human at this moment.
- (2) In case we detect humans with the bounding box in the next several frames as well we render 3D human
- (3) Return to (1).

3 EXPERIMENTAL RESULTS

We conduct an experiment using MHAD data [1]. We used the motion capture as input. After converting this to images, we obtain the 3D rendering of human as output. Some frames of resulted 3D human is depicted in the following figure. We attached the image file as the proof that the resulted actions are actually taken in the simulator.

For the MHAD data, the first problem was occurred in 10.6% and the second was 4.8%. The improvement by our algorithm was 59.4% and the second was 50.2%.

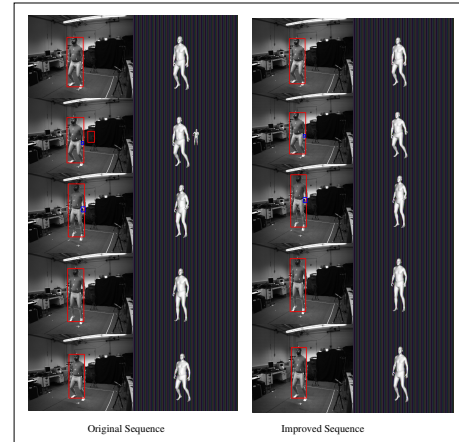


Figure 2: Activity simulation: Left is the rendering results by the original method and the right is the improved results. In the second row, the Right is the rendering results by our proposed method. In this simulation, we could eliminate the non-existent human and the sporadic ghost humans.

4 CONCLUSION

We present our new method to do a simulation based on the input signals. When we combine our previous method of translating between signals, with the motion rendering systems, we faced two problems which let the simulation ability decrease. We proposed two algorithms to increase the performance of this simulator in this preliminary work.

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